



UNIVERSITY
OF TRENTO - Italy

Dipartimento di Ingegneria e Scienza dell'Informazione

Know
dive



The Logics pyramid (HP2T)

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- **Recap**
- The DIKW pyramid
- The Knowledge pyramid
- The Logics pyramid
- Models
- World models
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- Language logics
- Informal-to-formal-to-formal-to-informal
- Key Notions

Four nested levels of mental activities (recall)

- **Perception.** From *reality* (i.e., *the real world*) to, e.g., the *images* of a “smiling face”, the “face of an old/young woman”, suitably composed into, e.g., the “**complex**” *images* of “a smiling face in a plug”, “an old woman looking down and left to something”. Images are (**complex**) **sensations** from reality. When we recognize a sensation previously perceived we come to the perception of a (**complex**) **percept**.
- **Conceptualization.** From, e.g., *images* to *words*, that is, an **alphabet**, ... **naming** what we perceive (e.g., the words: “face”, “old”, “young”, “woman”; “plug”), as we have perceived it before. In turn, these words are composed in **sentences** (e.g., “face of an old/young woman”, “an old woman looking down and left to something”) which describe what we perceive, as we have perceived it before. These words and compositions of words express (**complex**) **concepts** which allow to name (complex) percepts which occur multiple times. Thus for instance “cat” names all the times I have perceived the percept (sensation) of a cat.
- **Understanding.** From *words* and *sentences* denoting **concepts** to **assertions** (e.g., “face of an old/young woman”, “an old woman looking down and left to something”) which describe what is the case in the world, as we know it, and that integrate and extend what we know in case we did not know it. The words and sentence generated during conceptualization are **aligned** with what it is already known.
- **Reasoning:** From *assertions* to *propositions* stating the truth or falsity of assertions (e.g., “It is true that Fausto is the father of Benedetta”). Propositions are used by **reasoning** to derive **new propositions** / **assertions** which describe what **must** be the case as a **consequence** of what we know, where we have learned this type of consequence **in time**, and so on, recursively.

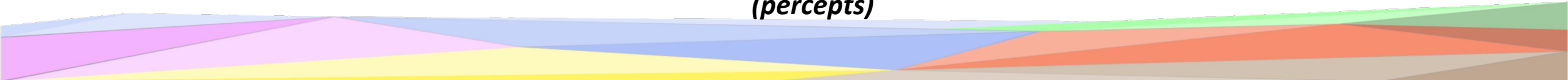
Example of reasoning

- FROM IF “(it is **true** that)” “it rains” **THEN** “(it is **true** that)” “the ground is wet” **AND** “(it is **true** that)” “it rains”
- DEDUCE “(it is **true** that)” “the ground is wet”

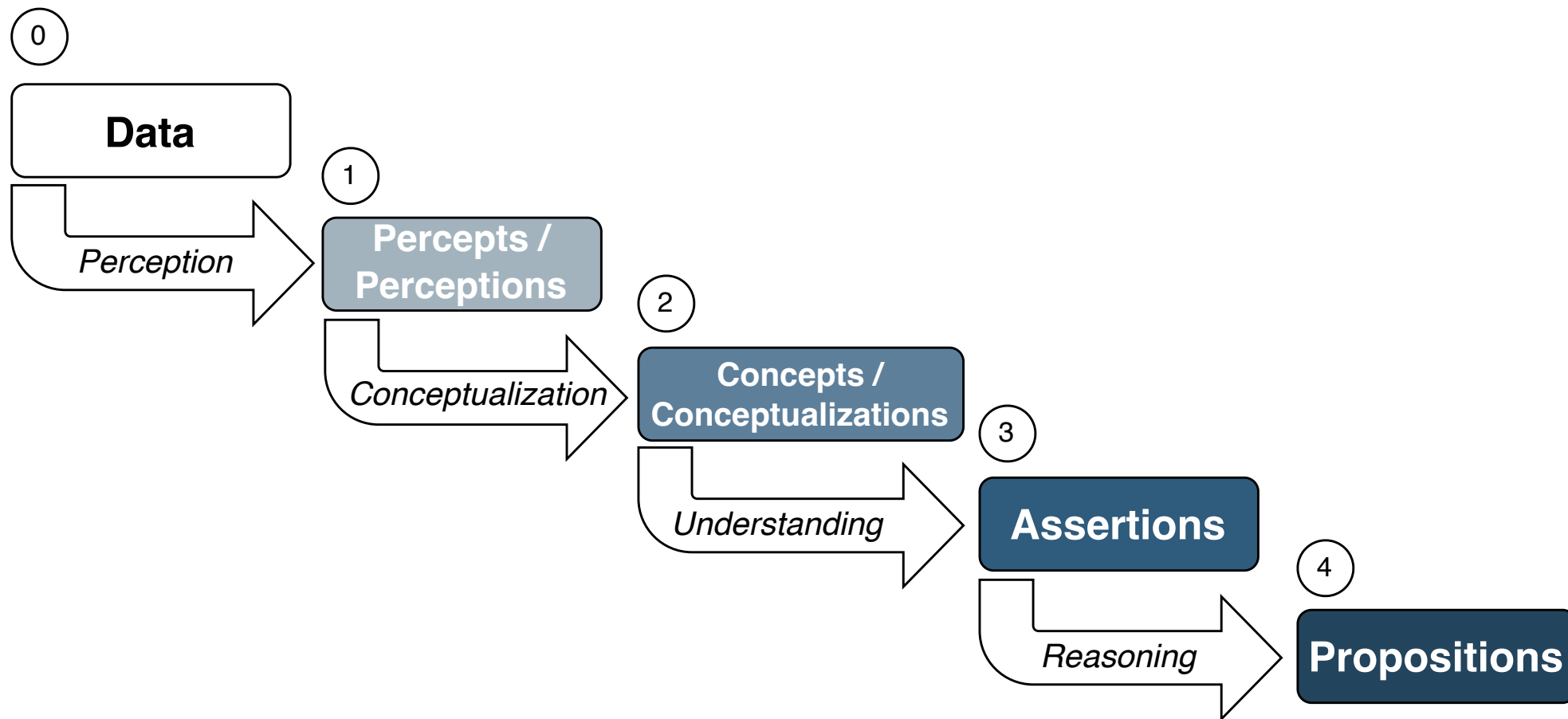
Four nested levels of mental representation (recall)

- **Percepts /perceptions.** From reality, the source of what we perceive, that is, **sensations**, expressed as, e.g., **images**, (e.g, the image of a person, the image of a cat), to **percepts**, e.g., the image of person recognized as such, and combinations of percepts into **complex percepts (perceptions)**, e.g., the image of a woman having, as **parts**, a face and a body. Complex percepts **depict** complex **mental representations** of reality. Example of a percept: “*smiling face*”. Example of a perception:
 - **Concepts/ conceptualizations.** From **percepts** and **complex percepts** to **concepts** and **complex concepts (conceptualizations)**, expressed using **words** and **complex combinations of words**, i.e., sentences. Concepts **name** the various kinds of **percepts**, while complex concepts **describe** complex **mental representations** of reality.
 - **Assertions.** From **(complex) concepts** to **properties, entities, entity types** ... composed to generate **assertions describing facts**, composed into **knowledge**. From combination of words which describe the world to their **alignment with the prior knowledge**, thus leading to the their **integration with and extension of the same**
 - **Propositions:** From **assertions** to **propositions** used for reasoning, thus enabling the extension of **existing knowledge** with **new knowledge**.
- ... and so on recursively in indefinite loop

perceive (percepts) – conceptualize (concepts) - understand (assertions)– reason (about proposition) – perceive (percepts)



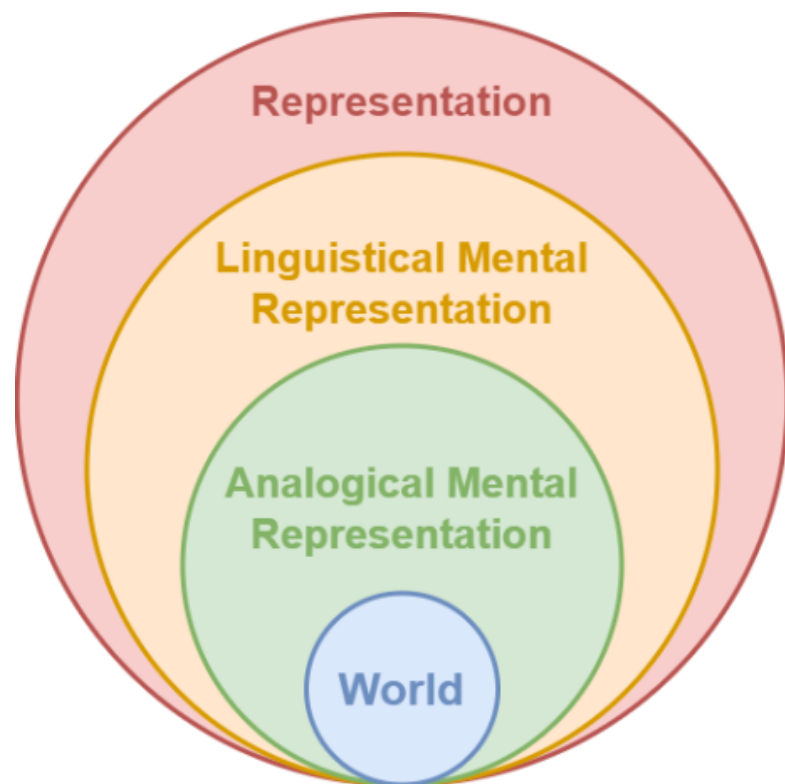
Four types of mental activities and representations



Representations

Two types of representation

- **Analogical representations**
 - Percepts, perceptions
- **Linguistic representations**
 - Concepts, conceptualizations
 - Assertions (entities, properties, etypes)
 - Propositions



We need to formalize all these types or representation and the process for generating them and using them when reasoning

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The DIKW pyramid – from Information Science



Ackoff offers the following definitions of data, information, knowledge and wisdom, and their associated transformation processes:

- Data are defined as symbols that represent properties of objects, events and their environment. They are the products of *observation*. But are of no use until they are in a useable (i.e. relevant) form. The difference between data and information is functional, not structural.
- Information is contained in descriptions, answers to questions that begin with such words as who, what, when and how many. Information systems generate, store, retrieve and process data. Information is inferred from data.
- Knowledge is know-how, and is what makes possible the transformation of information into instructions. Knowledge can be obtained either by transmission from another who has it, by instruction, or by extracting it from experience.
- Wisdom is the ability to increase effectiveness. Wisdom adds value, which requires the mental function that we call judgement. The ethical and aesthetic values that this implies are inherent to the actor and are unique and personal.

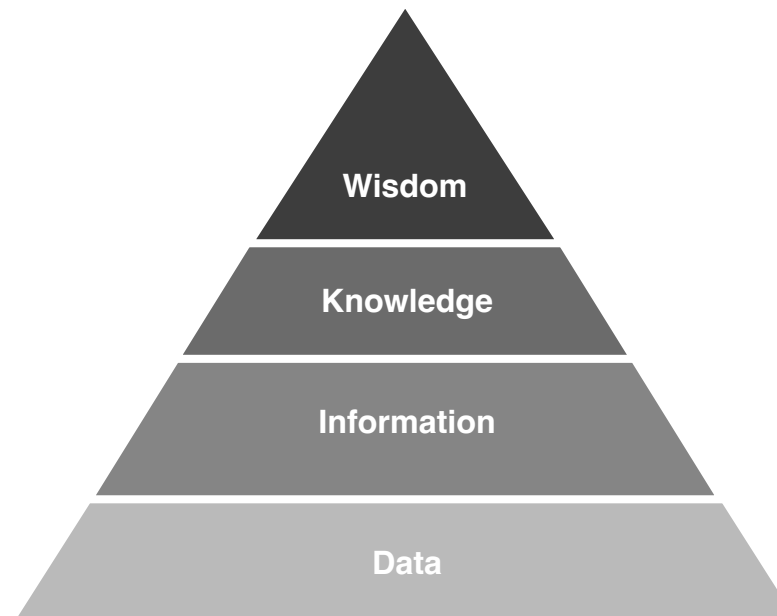
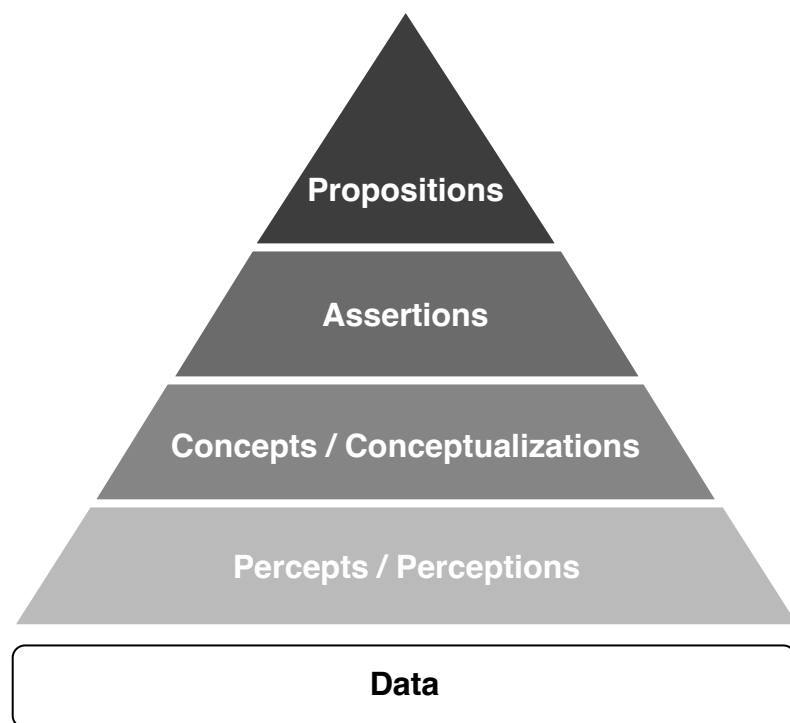
Ackoff, R. L. (1989). From data to wisdom. *Journal of applied systems analysis*, 16(1), 3-9.

Rowley, J. (2007). The wisdom hierarchy: representations of the DIKW hierarchy. *Journal of information science*, 33(2), 163-180.

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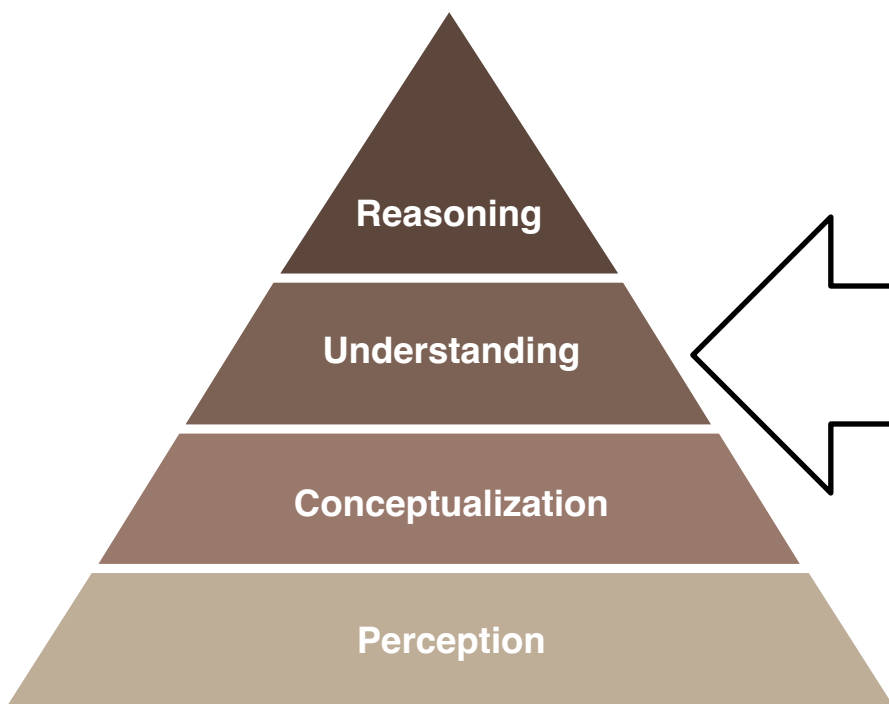
Representation pyramid vs. DIKW pyramid



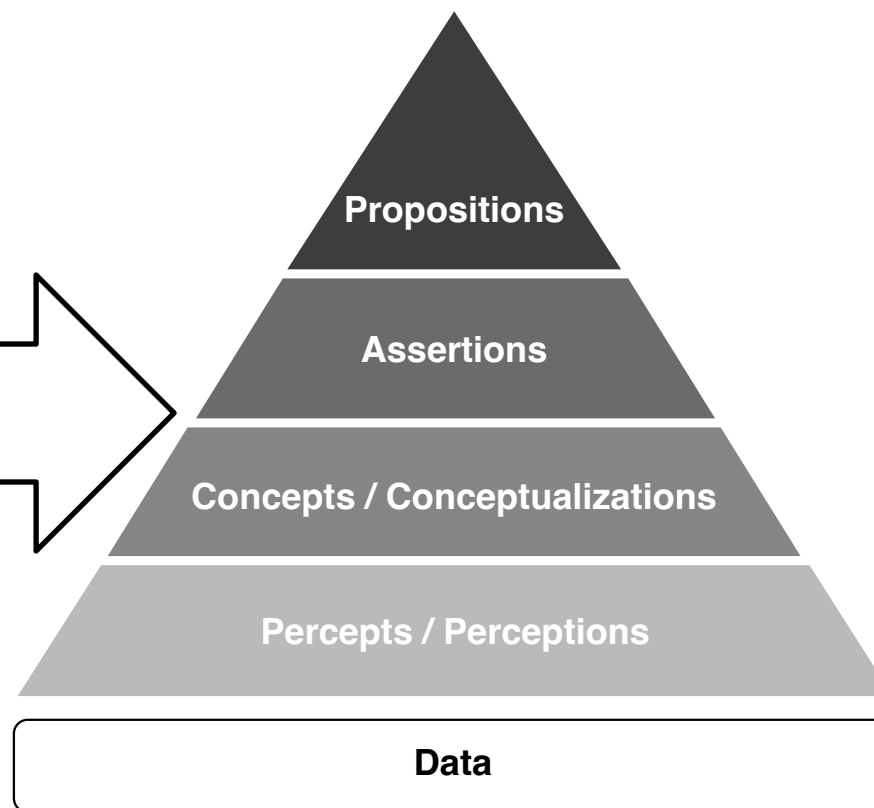
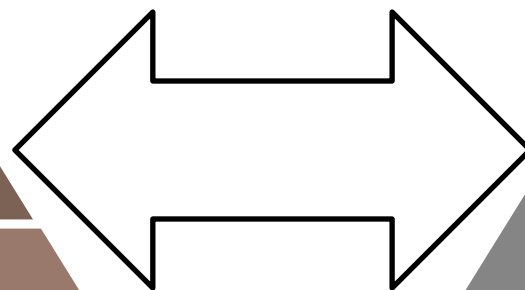
NOTE:
*the split of
data
(DIKW pyramid)
into
data and percepts
(representation
pyramid)*

The **representation pyramid** defines the 4+1 types and levels of **representation** and how they are generated, as function of the previous level

Activity pyramid vs. representation pyramid



The activity (PCUR) pyramid

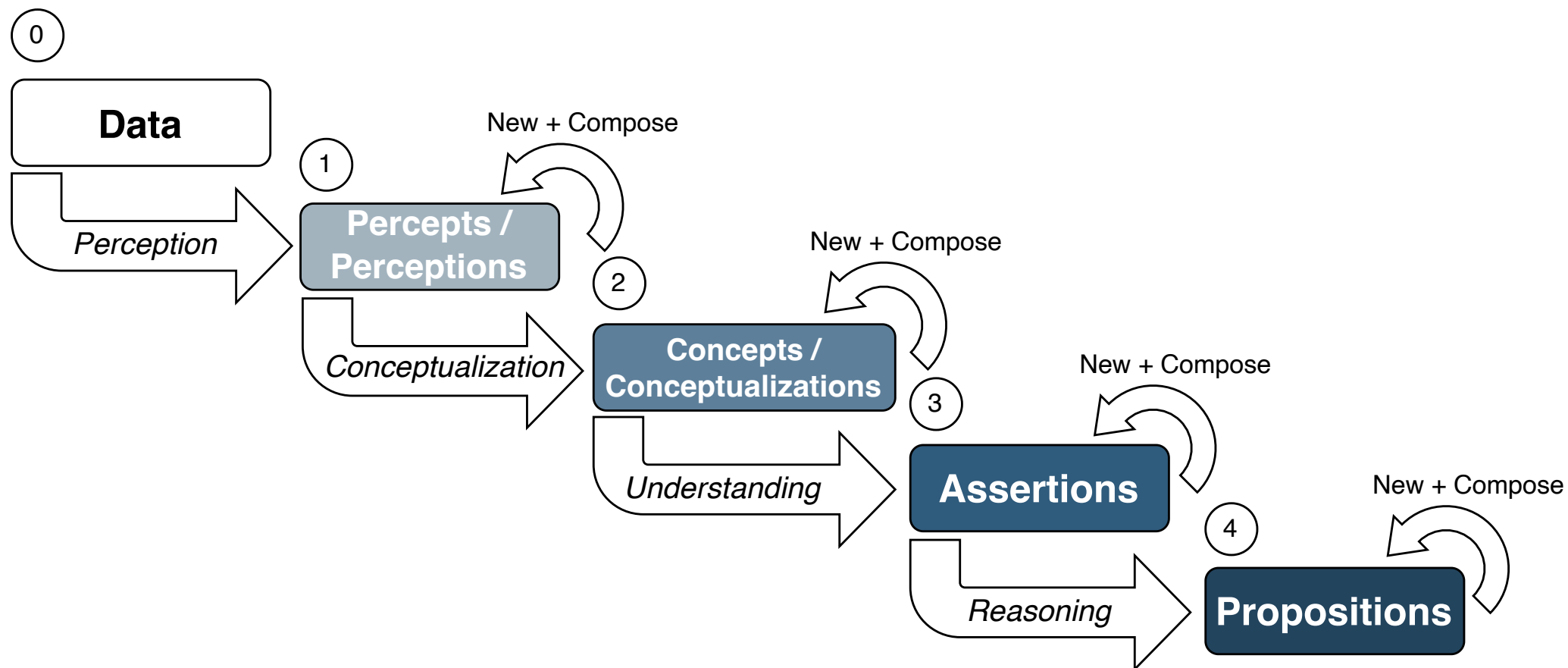


The representation (PCAP) pyramid

NOTE:
The two pyramids
are
NOT aligned

**Activity
generates
representation**

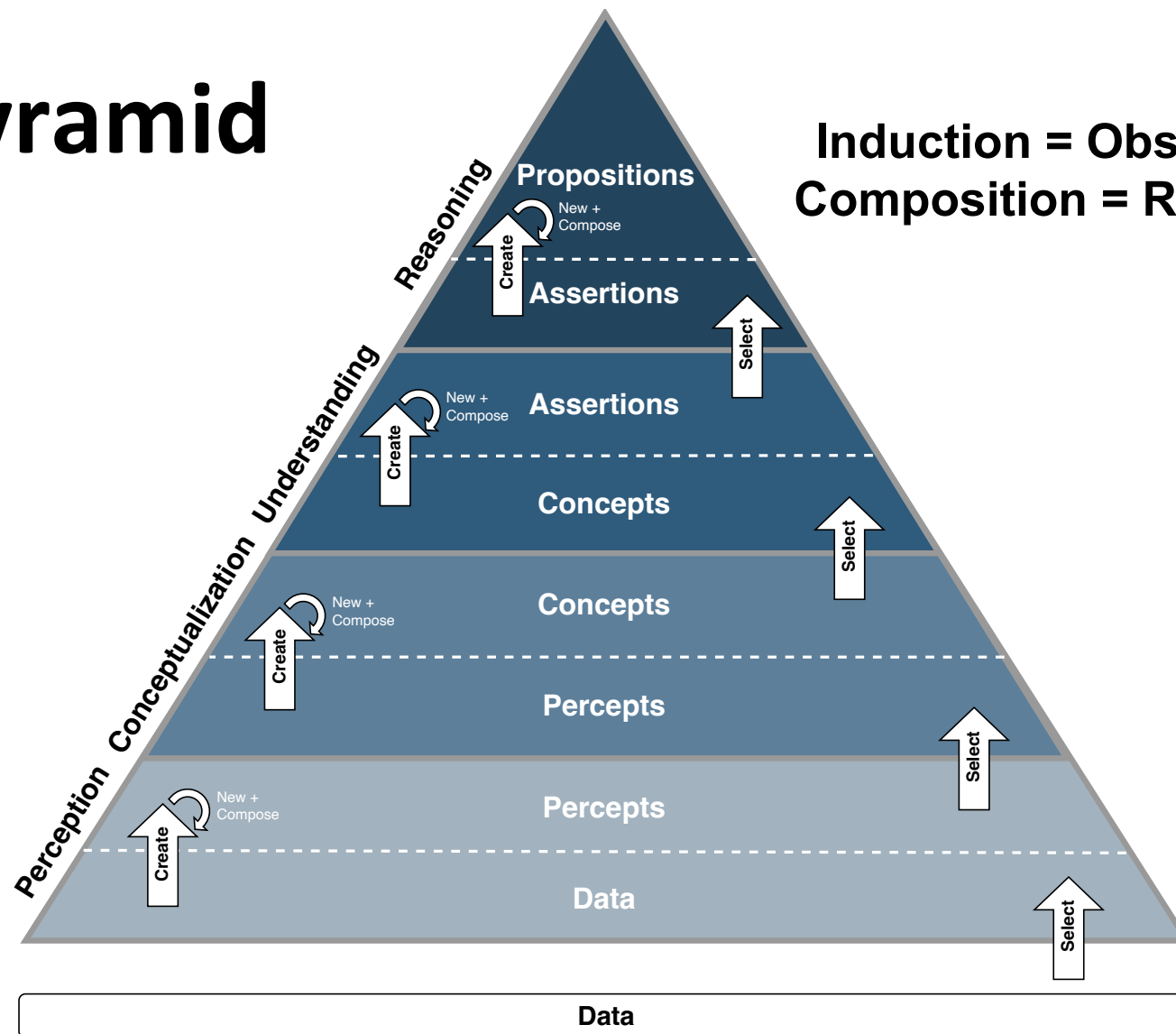
Mental representation and mental activities



The Knowledge Pyramid

Two steps:

1. **Create** a new «local» representation by **inducing** from the previous layer
2. **Create** a new «local» representation by **composing** what you have already created

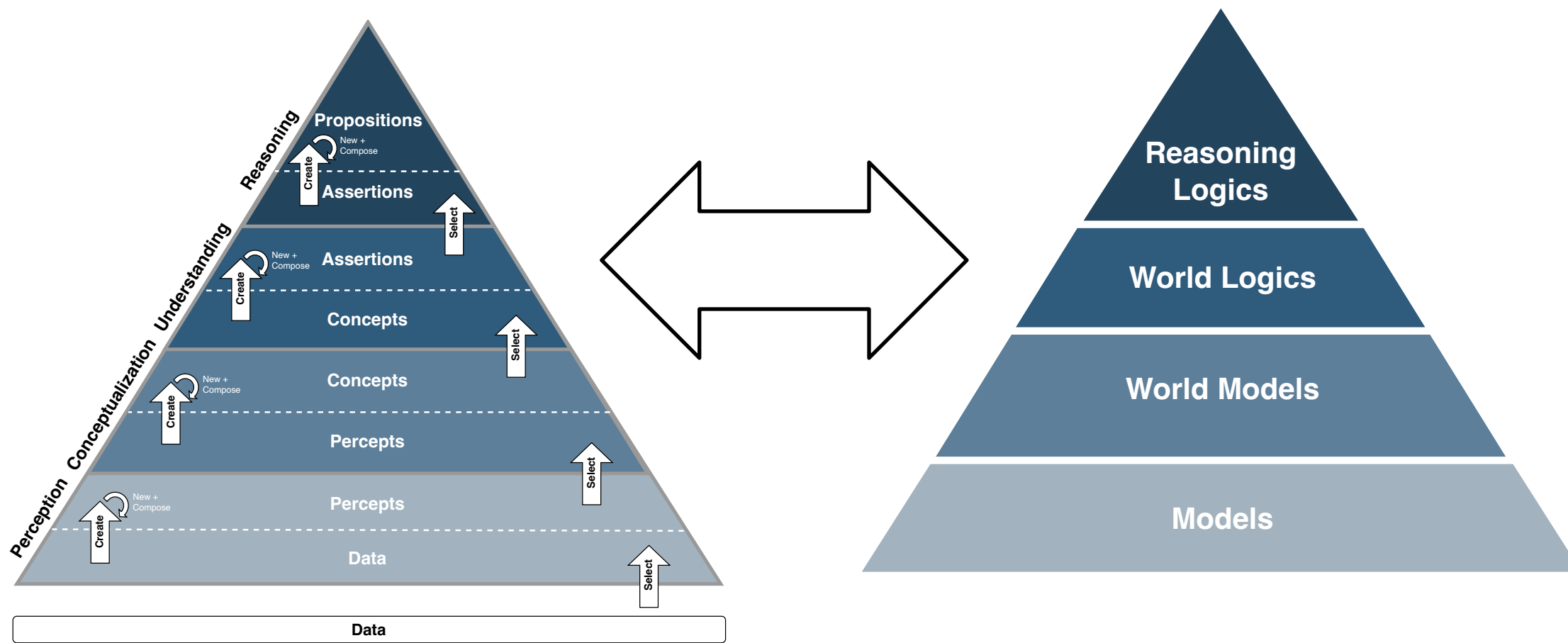


Induction = Observation
Composition = Reasoning

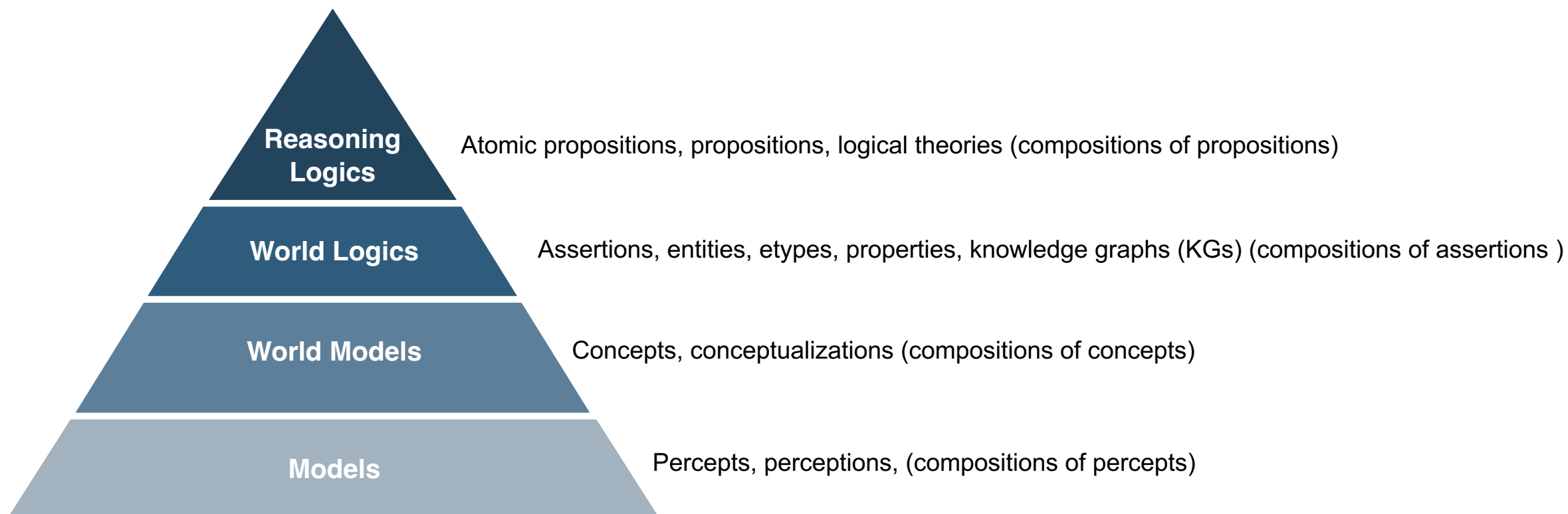
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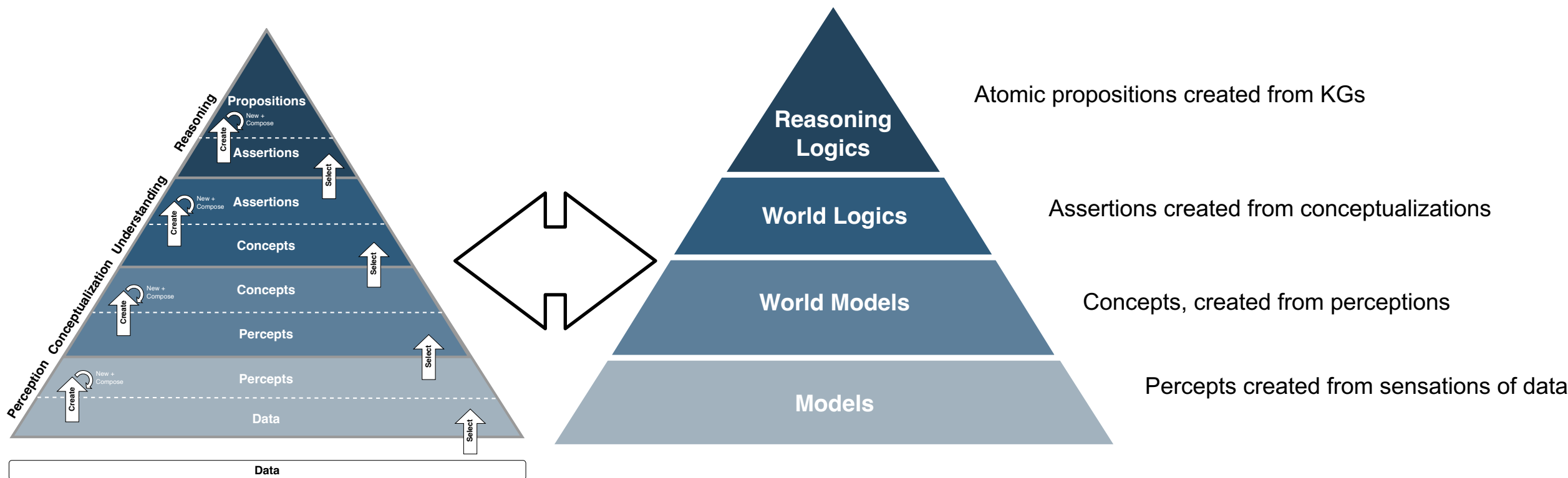
The knowledge Pyramid and the Logics pyramid



The logical representation of mental representations



The knowledge Pyramid and the Logics pyramid



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Models

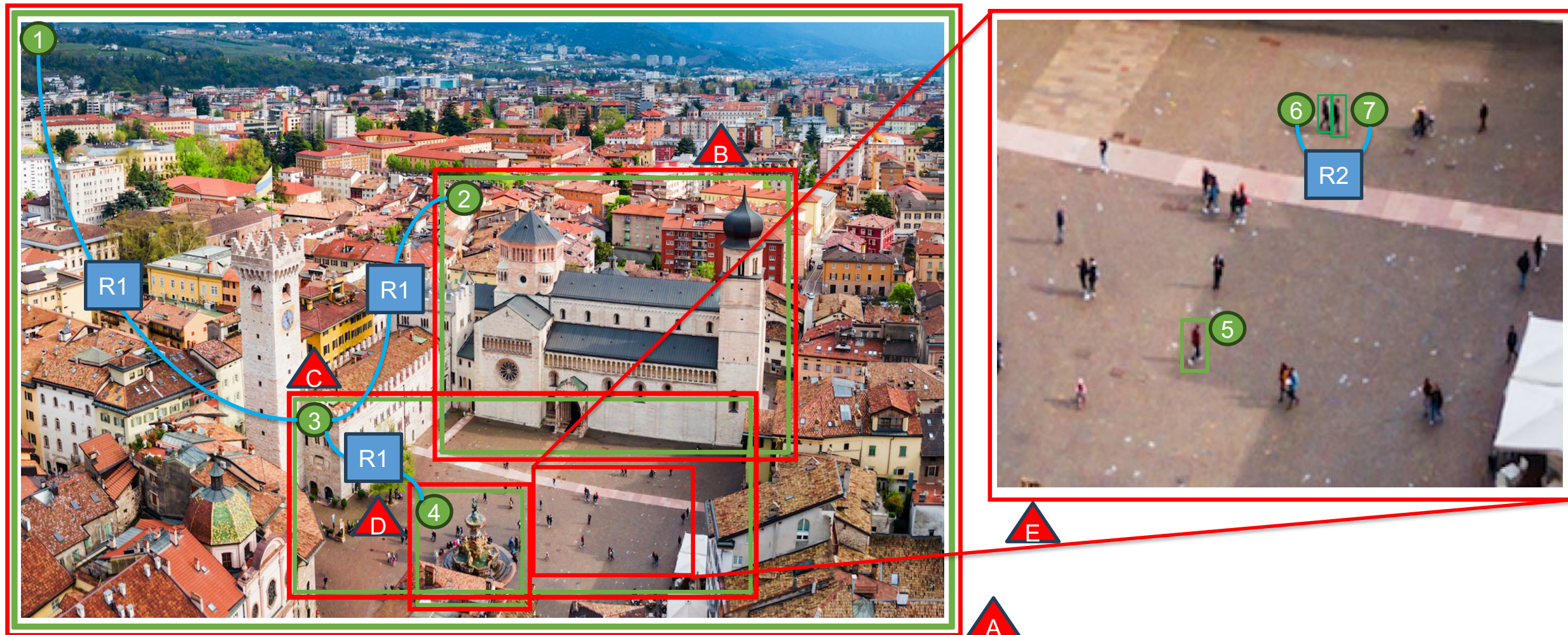
Intuition (model as composition of percepts and perceptions). A model is a **partial description** (and not a **depiction**) of an analogic representation, built as a complex **composition of percepts** (a **perceptpion**), where a **percept /perception** is a representation of a – possibly complex or abstract – element of the analogic representation being described.

Example (percepts represented in a model). Some examples are abstract representations of what we perceive: «a person», «a building», «a car», «the world», «walking», «eating», «being near», «beauty», «mind», «thought», «sleep»).

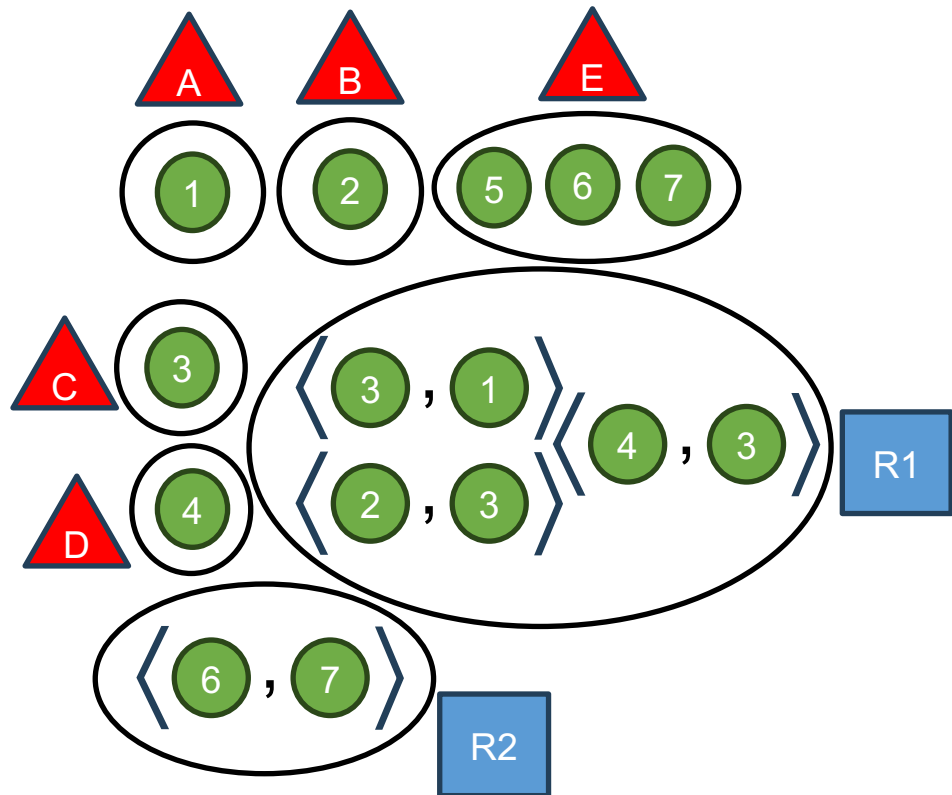
Example (perceptions represented in a model). Some examples are abstract representations of what we perceive: «a person with its face, arms and body», «a building with windows and doors», «walking as a sequence of steps and arm movement», «a person being near to another person», «thought as a series of component thoughts»).

Intuition (model). A model is a representation which allows to describe in a not ambiguous formalism what has been perceived without committing to a specific language. It is an a-linguistic description of the world

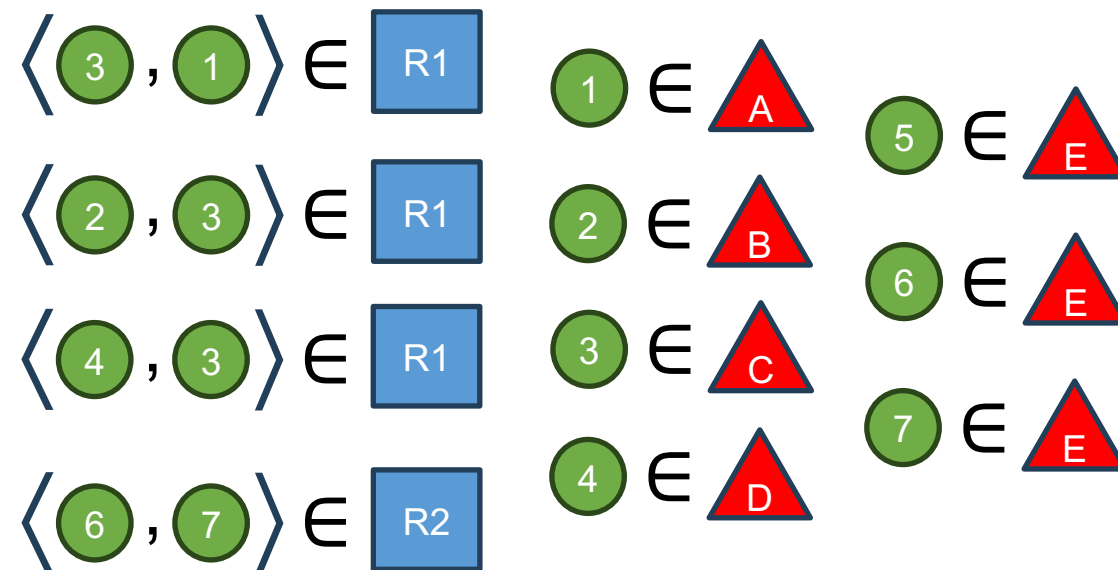
An analogic representation, as we perceive it



Models – Terminology



Percepts



Perceptions

(percepts composed in complex percepts)

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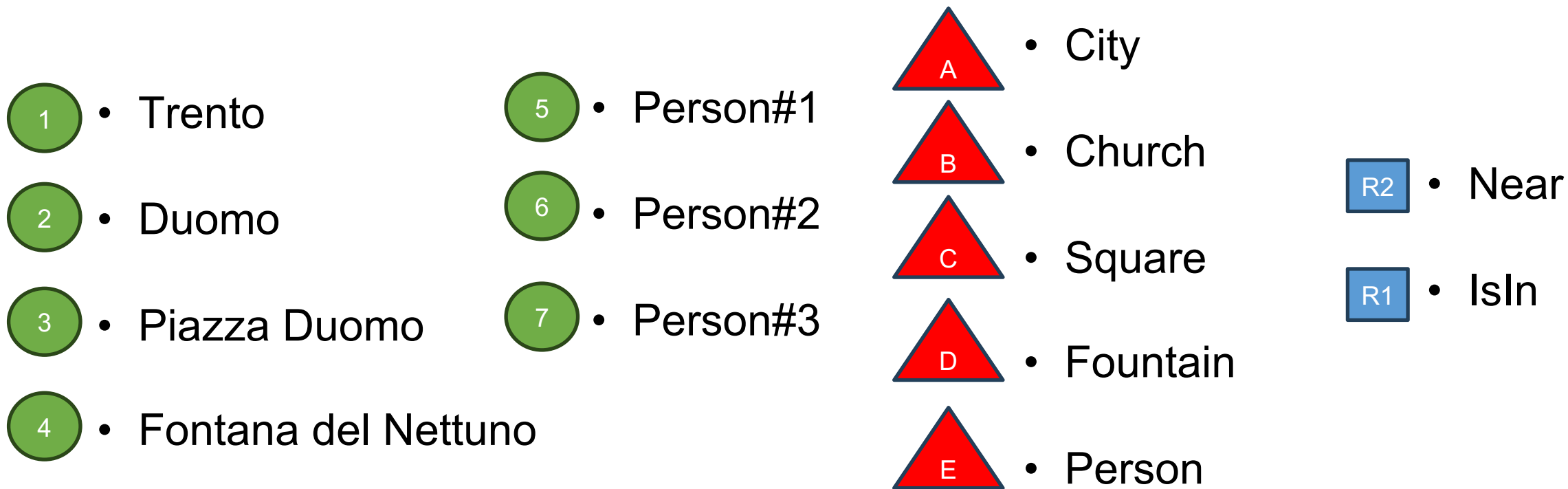
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World models

Assume you have chosen a model of the world, also called the **reference model**. A world model is:

- The reference model
- A language which allows to **name (denote)** (some of) the percepts and perceptions of the reference model
- The mapping from names to percepts is not ambiguous. A name cannot denote more than one percept, but a percept can have multiple names
- The names of perceptions are composed from names of percepts following a one-to-one to structure of perceptions, starting from the names of the constituent percepts

The meaning of concepts from Wordnet@Trento



The meaning of concepts is built by «attaching them» (so called **interpretation function**, see later) to the percepts of the reference model

The meaning of conceptualizations from Wordnet@Trento

① ∈ △_A • City(Trento)

② ∈ △_B • Church(Duomo)

③ ∈ △_C • Square(Piazza Duomo)

④ ∈ △_D • Fountain(Fontana del Nettuno)

⟨②, ③⟩ ∈ □_{R1} • IsIn(Duomo, Piazza Duomo)

⟨④, ③⟩ ∈ □_{R1} • IsIn(Fontana del Nettuno, Piazza Duomo)

⟨③, ①⟩ ∈ □_{R1} • IsIn(Piazza Duomo, Trento)

⑤ ∈ △_E • Person(Person#1)

⑥ ∈ △_E • Person(Person#2)

⑦ ∈ △_E • Person(Person#3)

⟨⑥, ⑦⟩ ∈ □_{R2} • Near(Person#2, Person#3)

NOTE: the meaning of conceptualizations is compositional on the meaning of concepts

Models and world models - terminology

We use the following terminology:

- A **perception** is a percept composed out of one more percepts
- An **atomic percept**, or simply a **percept**, is a percept which is not a fact
- The name of an atomic percept is called a **concept**
- The set of concepts is an **alphabet**
- The name of a perception is a **conceptualization**, sometimes called also a (linguistic, model) **description**
- The set of rules which define how conceptions are build from concepts is a **grammar**, also called **syntax**
- A conception can be named with a concept (**definition**).

Intuition (world model). A world model is a representation of the world which provides a language which allows to unambiguously name (denote) the concepts and conceptualizations naming the percepts and perceptions of the reference model

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World Logics

A World logic is

- A world model, also called the **reference world model**
- The possibility to derive whether a certain assertion is a consequence of one or more assertions which are known to be the case. One such operation is called an **inference rule**.
- The possibility to compose multiple inference rules into a so-called **derivation** (also called **deduction**)
- The possibility to **choose** in case multiple inference rules are applicable.

Terminology (Assertion). An **assertion** is a **conceptualization** which is known to be the case. Assertions, similarly to conceptualizations, are composed out of **etypes**, **entities**, and **properties** each mapping to a different type of concept

Intuition (World Logic). A World Logic is a representation of the world which allows to reason about what is known of the reference world model, that is, to deduce assertions which are implied by the known assertions.

World models and world logics - terminology

Concepts:

- City
- Church
- Square
- Fountain
- Trento
- Duomo
- Piazza Duomo
- Fontana del Nettuno
- IsIn
- Person
- Near
- Person#1
- Person#2
- Person#3

Etypes: City, Church, Square, Fountain, Person

Relations: IsIn, Near

Entities: Trento, Duomo, Piazza Duomo, Fontana del Nettuno, Person#1, Person#2, Person#3

Conceptualizations / Assertions:

- City(Trento)
- Church(Duomo)
- Square(Piazza Duomo)
- Fountain(Fontana del Nettuno)
- IsIn(Duomo, Piazza Duomo)
- IsIn(Fontana del Nettuno, Piazza Duomo)
- IsIn(Piazza Duomo, Trento)
- Person(Person#1)
- Person(Person#2)
- Person(Person#3)
- Near(Person#2, Person#3)

World logics – reasoning

Entity types (etypes):

- Church
- Building

Entities:

- Duomo

Properties:

- hasBellTower

Assertions:

- Church(Duomo)
- Church $\sqsubseteq$ Building $\sqcap$ hasBellTower

New assertions:

- Building(Duomo)
- hasBellTower(Duomo)

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Language Logics

A language logic is

- A world logic, also called the **reference world logic**
- The possibility to derive whether a certain proposition is a consequence of one or more propositions. One such operation is called a **(logical) inference rule**.
- The possibility to compose multiple inference rules into a so-called **(logical) derivation** (also called **(logical) deduction**)
- The possibility to **choose** in case multiple inference rules are applicable.

Terminology (Atomic proposition, proposition). An **atomic proposition** is a judgement (True / False) on whether an assertion (a KG) is a representation of the reference world model. A **proposition** is a composition of atomic propositions.

Intuition (Language Logic). A Language Logic is a representation of the world which allows to reason about what is NOT known of the reference world model, that is, to deduce propositions which are implied by the known propositions. These propositions define precisely what is known and what is NOT known of the reference world model.

World logics and Language logics - terminology

Assertion:

- $\text{City}(\text{Trento})$

Propositions:

- $\text{CityTrento} = \text{"It is True that Trento is a city"}$
- $A = \text{"It is True that Trento is a city"}$
- $\text{NCityTrento} = \text{"It is False that Trento is a city"}$
- $B = \text{"It is False that Trento is a city"}$
- $A \supset \text{not } B$
- $\text{NCityTrento} \supset \text{not NCityTrento}$
- $\text{City}(\text{Trento}) = \text{"The etype" City "is True of" Trento}$

NOTES:

1. Propositions are judgements on the truth of propositions
2. The specific judgments are up to the choice of logic designer, as long as the mapping is done consistently across all assertions
3. The first set of judgements are expressed in the Logic of Propositions (LoP often called PL).
4. The last judgement, much more expressive, is expressed in the Logic of Interactions (LoI, often call FOL, First order Logic)
5. In Language Logics, the specific (form of the) proposition is irrelevant to the reasoning. **What is relevant is only its Truth/Falsity**. Language Logics are detached from the reality they describe

World Logic

- Person(Person#1)
- City(Trento)
- livesIn(Person#1, Trento)

Reminder: *The specific linguistic formulation is only mnemonic to the reader but it plays NO role in the reasoning pattern*

Language Logic

$$\forall x \forall y. \text{livesIn}(x,y) \supset \text{studiesIn}(x,y) \vee \text{worksIn}(x,y)$$

FROM livesIn(Person#1, Trento) $\supset$ studiesIn(Person#1, Trento) $\vee$
worksIn(Person#1, Trento)

AND livesIn(Person#1, Trento)

DEDUCE studiesIn(Person #1, Trento) $\vee$ worksIn(Person #1, Trento)

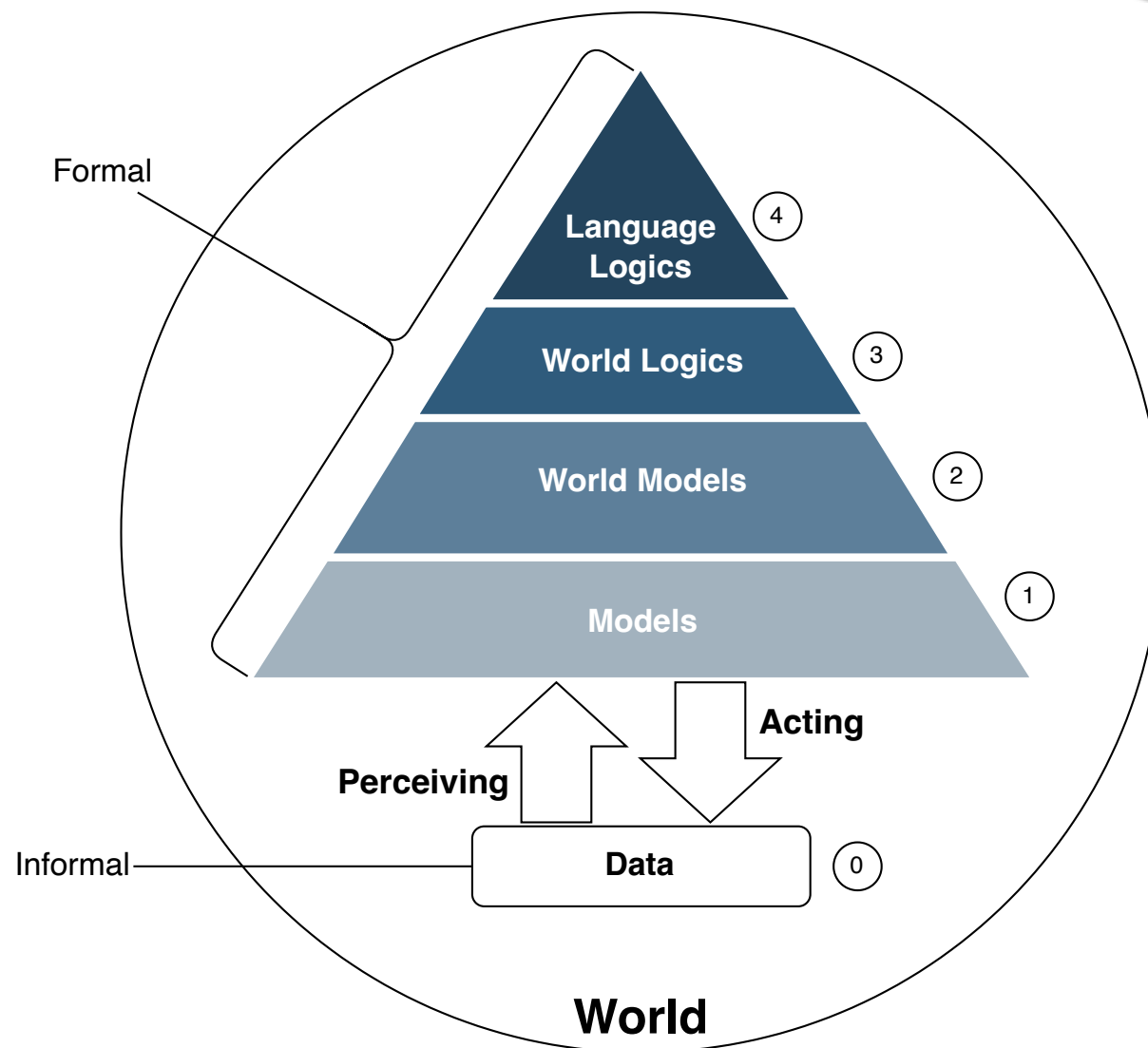
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Informal-to-formal-to formal-to-informal

Key observation:

- We can guarantee correctness (and also completeness) of the reasoning – *with respect the intended reference model* – while in the process of **reasoning** inside the logic pyramid (formal-to-formal)
- We cannot guarantee correctness (nor completeness) in the process of **perception** (informal-to-formal) nor when in the process of **acting** (formal-to-informal)



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Key notions

- The DIKW pyramid
- The Representation pyramid
- The Activity pyramid
- The Knowledge pyramid
- The Logics pyramid
- Models, percepts, perceptions
- World models, concept conceptions
- World logics, assertions, etypes, entities, properties, KGs
- Language logics, atomic propositions, propositions
- Informal-to-formal-to-formal-to-informal (I2F, F2F, F2I, I2I)

On the left, are all the key notions of the course

The course will cover the most important world and language logics which allow to reason about the foundational types of world models (LoE,LoC,LoD, LoDe, LoP, Lol) and the related I2F2F2I issues



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The Logics pyramid (HP2T)